

Cloudflare Radar 2024 Year in Review

A data-driven analysis of global Internet traffic,
adoption trends, connectivity disruptions, and security
threats.

17% Global Traffic Growth and Starlink's 3.3x Surge

Global Internet Traffic

- Grew by 17% in 2024, continuing a strong upward trend.
- Driven by increasing reliance on communication, commerce, and entertainment.

SpaceX Starlink

- Traffic saw dramatic 3.3x growth globally.
- Continues expanding satellite Internet to previously underserved locations.



Google Leads Top 10 Services as Generative AI Rises

Most Popular Services

- Google maintains the #1 overall position for the second consecutive year.
- Facebook, Apple, and TikTok follow closely in the top 4.

Category Shifts

- Generative AI services continued their rapid, meteoric rise throughout 2024.
- The Metaverse receded into the background, losing traffic share.

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

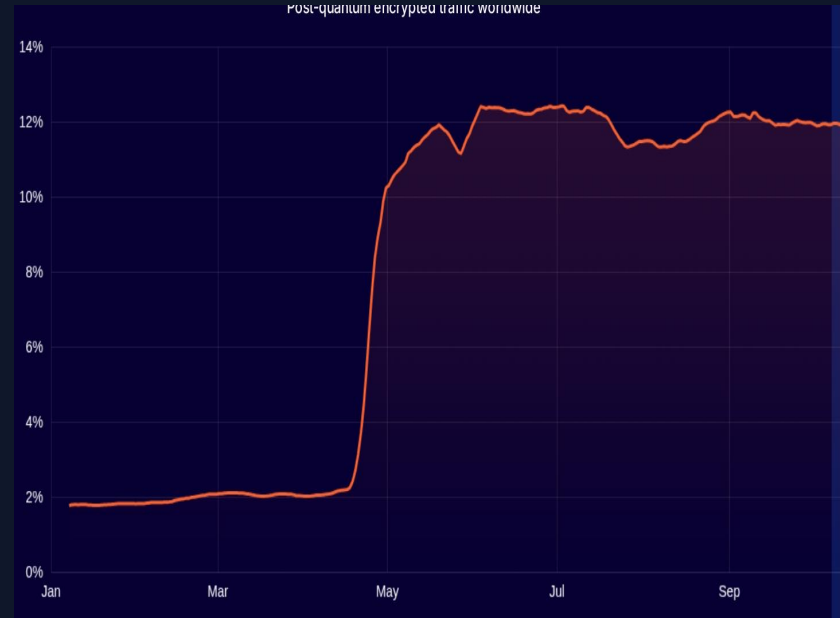
Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

Post-Quantum Cryptography

- Adoption skyrocketed from 2% to 13% after Chrome 124 enabled it by default.
- Firefox rolling out defaults; Safari in initial testing.

Protocol & Language Adoption

- HTTP/3: Now accounts for 21% of all traffic, providing faster, more secure connections via QUIC.
- API Clients: Go is the #1 API client language (12% of automated requests), followed by Node.js and Python (9.6%).
- IPv6: Reached 28% of dual-stack traffic.



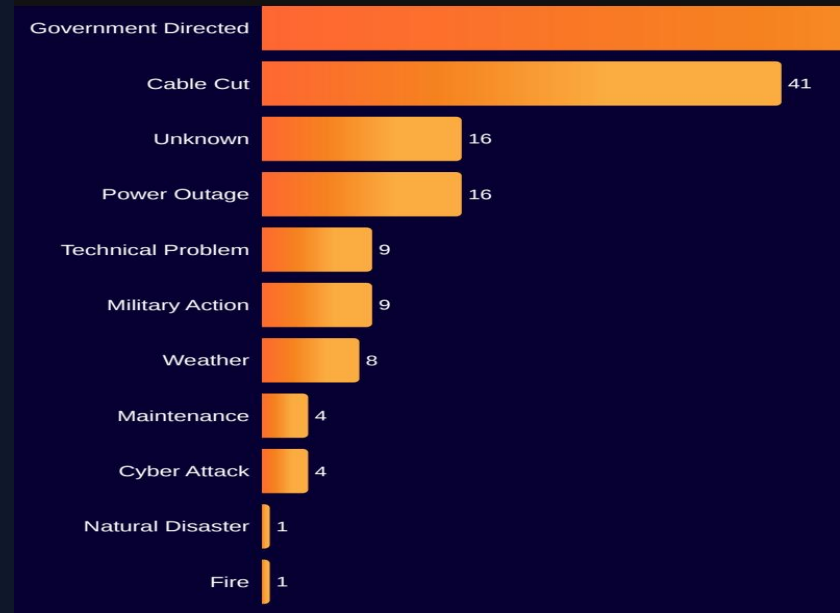
225 Major Disruptions: Government Shutdowns Lead Outage Causes

Global Outages in 2024

- Cloudflare tracked 225 major Internet disruptions worldwide.

Primary Causes

- Government Directed: 116+ incidents (Over 50% of all major outages).
- Cable Cuts: 41 incidents.
- Unknown & Power Outages: 16 incidents each.
- Other factors included military actions, technical problems, and weather.



6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

Network & Application Security

- Overall Mitigation: Cloudflare mitigated 6.5% of all processed traffic.
- Top Target: The Gambling and Games sector faced the most attacks.
- Bot Concentration: 68% of global bot traffic originated from just 10 countries.
- Vulnerability Persistence: Log4j continues to be targeted over two years after disclosure.

Email & Routing Threats

- Malicious Emails: 4.3% of processed emails were malicious.
- Deceptive Links: Contained within 43% of malicious emails.
- Top TLD Threat: .bar originated the largest share of spam and malicious emails.
- Routing Security: RPKI valid routes increased 6.4%, improving baseline Internet routing.

Five Takeaways Shaping the 2025 Internet Outlook

1. Sustained Traffic Growth

17% YOY growth and satellite expansion (Starlink) prove demand continues to accelerate.

2. Post-Quantum Era Begins

Default enablements in Chrome pushed TLS 1.3 post-quantum encryption to 13%, setting a new standard.

3. Connectivity is Fragile

Government-directed shutdowns accounted for over half of the 225 major disruptions globally.

4. Persistent Security Debt

6.5% mitigation rates and ongoing Log4j exploitation show the long tail of known vulnerabilities.